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Dear Annalisa Crannell,

I am submitting my article *The Continuous-Coefficient Jensen Equation: How Real Mixing Weights Retire Regularity* for consideration by *The American Mathematical Monthly*. The work is original, unpublished, and not under consideration elsewhere.

At the center of the paper is a small but decisive reversal. A century of analysis has taught readers to expect regularity hypotheses— continuity, measurability, monotonicity, boundedness—before Jensen's equation yields an affine conclusion. That instinct is exactly right for rational weights and exactly wrong once the mixing weight ranges over the full interval $[0, 1]$. In that setting, a single endpoint substitution forces the solution onto its chord, with no regularity hypothesis at all.

The manuscript treats that one-line observation as the entrance to a larger landscape. It draws a sharp boundary between settings in which the classical hypotheses still do real work and settings in which they have quietly become vestigial; explains why Hamel pathology survives the rational tests yet fails at irrational weights; and gives the reader a compact diagnostic for telling those situations apart. The aim is not just to settle the equation, but to show why one extra degree of mixing changes the logic so completely.

This seems well suited to *Monthly*. The proof is brief and transparent, the central question is easy to state, and the payoff is wider than the calculation itself: a familiar functional equation turns out to separate several neighboring literatures in an unexpectedly clean way. The article is written for readers who enjoy a result that can be grasped quickly and then keeps unfolding.

I have no conflicts of interest to disclose.

Please address correspondence to me at aelouafiq@sqli.com. Thank you for your consideration.

Sincerely,

Ali Elouafiq